CLIMATE CHANGE IMPACT ON BANGALORE

1 ABSTRACT(173 words)

Bangalore once known as a “garden city” showing a consistent cool environment is now shifting to a milder environment with rising in its mean temperatures showing a more “oven city” environment with extreme rainfall also causing many foods. This paper focuses and analyzes the temperature trends of Bangalore mainly focusing on the changes in annual rainfall of Bangalore, loss of green cover and water bodies . The paper also aims to show the impact this has on the locals and develop ideas for people to adapt and a roadmap for urban planning .The data visualisation charts provided in the paper shows the= annual maximum and average temperature as well as annual rainfall and distribution of rainfall through the different months to conclude the air quality difference between summers and winters .The paper is also provided with an influence of interviews with two old local citizens of Bangalore. In the end the paper intends to educate readers about environmental sustainable urban planning , improvement in the air quality and the heat-reduction measures to be taken.

2 Introduction (264 words)

Bangalore has always been admired for its gentle weather, steady seasons, and green landscapes that once was named as the “Garden City.” For many years, the city enjoyed a predictable climate that kept its lakes alive, supported its biodiversity, and made it one of the most comfortable places to live in India. But over the last two decades, this familiar climate has changed noticeably. Summers feel hotter, the monsoon arrives uncertain , and water has become more scarce . Floods during heavy rain, rising temperatures, and declining groundwater levels show that Bangalore is no longer experiencing the climate it once had . These changes hit Bangalore harder than many other cities because of its growth rate . Expanding roads, buildings, and reduced trees catch more heat, creating strong UHI(urban heat island) effects. Bangalore now experiences floods, which were not common before , because there are fewer lakes and wetlands to store the rainwater . Together, these drawbacks make Bangalore more at risk from climate change .By examining temperature patterns, annual rainfall , and flood related issues, this study seeks to comprehend how Bangalore's climate has changed between 2000 and 2024. Locals' observations and experiences are also included. This study's primary questions are: How precisely has Bangalore's climate changed? *Why is the city so sensitive to these changes? And how are people experiencing the impact in their daily lives?* Studying climate change from the eyes of a single city matters because this is where people feel the effect most immediately . Understanding how Bangalore's climate is shifting can help in better planning and more sustainable development in the near future**.**

3 Background:bangalore’s climate & urban growth (423 words)

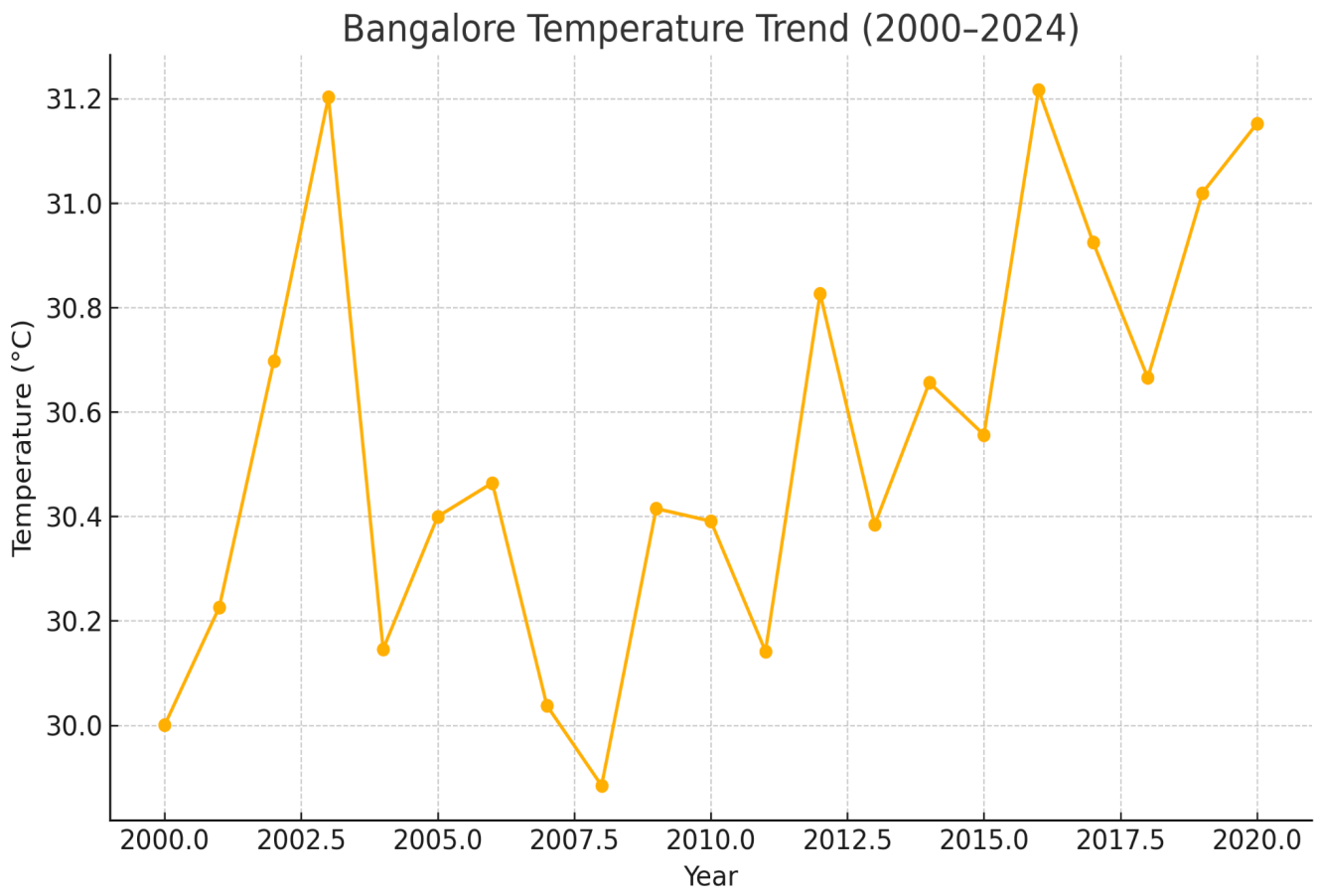
For a large period of time , Bangalore was known for having a climate that, in comparison to most Indian cities, seemed almost unreal - cool mornings, mild afternoons, and breezy evenings that remained comfortable throughout the year . Because of its landscape having a higher elevation on a decan plateau the temperatures were naturally lower there. Bangalore's distinct microclimate was there because of the city's chain of lakes, open fields, and tree filled streets. As a result, the city had its own local weather

pattern that remained moderate even when the surrounding areas were blistering . One of the main reasons people chose to live, work, and retire here was the climate.However, Bangalore's climate began changing significantly as the city expanded. The fast expansion of the IT sector in areas such as Whitefield , Electronic City, and Outer Ring Road brings in a lot of the population searching for jobs and education. this migration came along with massive construction of tech parks, gated communities, flyovers, and malls which is rapidly reshaping the city's landscape. The population blew up , and with it came the demand for housing, transport, and infrastructure.As nature and trees were replaced by dense construction, the city started heating up. Materials like concrete, glass, and asphalt trap heat way more than soil or water . The constructed buildings absorb heat during the day and release it slowly during the night , causing the temperatures to remain high This phenomenon is known as the UHI(Urban heat island) , it is now strongly visible in many parts of Bangalore. Areas with high traffic and fewer trees, developed parts of Bangalore usually suffers with hotter weather than in suburbs. Urbanisation has also influenced the water cycle. Before climate change the water bodies, wetlands, and open spaces allowed rainwater to naturally immerse into the earth. Today, the city is covered by hard surfaces - roads and rooftops which block the water from absorption. This has created an uneven water distribution where heavy rains quickly lead to floods while there is very less groundwater the level of which keeps dropping. Lakes that used to store and keep water have either shrunk or disappeared, removing the natural ways which kept the city's climate stable.On top of these challenges , global warming is also increasing temperatures everywhere, making Bangalore's climate shift even more evident . The combination of fast urban growth and global climate change has deeply reshaped what was once a stable, cool climate. Today, Bangalore's weather shows the issues of modern urbanisation and the global environmental changes

4 Rising temperatures - from "garden city" to "oven city"(402 words)

One of the most visible signs of climate change in Bangalore is the way the city has slowly but not willingly warmed over the years. According to IMD data, the temperatures have been rising slowly. Reports from the 1990s show summer afternoons usually peaking around 28–30°C, a temperature which justifies its reputation for its cool weather . Today, however, many summer days easily cross 34–36°C, and some even go beyond that . What is even more shocking is how hot the nights have become Instead of the cool17–19°C that locals once were staying in, night temperatures now often staying above 22–24°C, making the city feel warm even after the sun sets down.Heat waves, which were very rarely seen in Bangalore because of its landscape and greenery, have become increasingly common. The warm days that used to pass quickly now turning into long, exhausting days of heat. Many older locals say that the city simply doesn't "rest" the way it used to which shows the change in climate. Comparing the Bangalore of the 1990**s** with the **2020s** shows how the climate has changed. Summers now feel longer, winters are now also shifting and becoming milder, and shifts in temperature happen far more instantly . The "Garden City" climate that residents grew up with has slowly faded away, replaced by a new hotter climate , a more unpredictable version of the city. This shift becomes very evident when looking at the temperature trend chart in this research. The line gradually rises, reflecting what people feel on the ground, which is that the city is heating up.Two major scientific factors give evidence of this change. The first is the **greenhouse effect**, which is warming the whole planet up . As carbon dioxide and other greenhouse gases build up in the atmosphere, more heat gets

trapped , raising temperatures everywhere — including Bangalore.The second factor is the **UHI (Urban heat island) effect**. As Bangalore has grown, concrete and glass have been replacing trees, soil, and open spaces. These materials absorb heat during the day and release it slowly at night, making the city stay warm for longer than before. With fewer trees available to provide shade and cooling.Together, global warming and the UHI effect have transformed Bangalore's once cool climate into one that feels far more hot and harsh , a shift that many residents describe as the city turning from "Garden City" into "Oven City."



5 changing rainfall patterns & increased flooding (362 words)

Another observation which shows that the Bangalore climate has changed is the shift in the rainfall pattern . The city used to experience a calm and reliable monsoon, with rain all across the season. Older locals remember light rains that came daily, giving time to the lakes to fill up slowly and the ground to absorb water naturally. But now, rainfall has become far more uneven. Instead of being spread out over the whole season, Bangalore now gets sudden, heavy rain that

feels almost like a cloudburst. *These quick, heavy bursts of rain release so much water so fast that the drainage system fails to co-operate.* This change in rainfall behaviour is the cause for many major floods especially in 2005, 2022, and 2024. These events were not just heavy rain ,they were a time where neighbourhoods, roads, and even huge buildings were immersed within minutes. Viral videos from the 2022 floods, for example, showed cars floating in plateau areas , something few people could have imagined a decade earlier.The rainfall distribution chart included in this research shows this shift clearly. While annual rainfall might not look totally different, the *pattern* has changed dramatically. Now there is more and heavy rainfall in fewer days meaning the city receives the same amount of water but in a much harsher and destructive way.Several water system issues worsen the situation. Over the years, many stormwater drains (rajakaluves) have been narrowed or built over because of construction and development of new buildings . Lakes, which were used as natural storage spaces for water , have been blocked, reduced, or separated from each other. Due to this, rainwater that was able to easily flow now has nowhere to go. At the same time, the construction of roads and concrete surfaces means the ground absorbs very less water than before. This leads to faster and higher chances of flooding, and slower recovery after storms.To conclude, the combination of heavy rainfall, weak water systems, and rapid urbanisation has made Bangalore far more at risk of flooding than ever before. What used to be the “monsoon season” has turned into a cycle of unpredictable rain and frequent flooding .

6 Interview findings (**Two Senior Citizens of Bangalore**)

6.1 ramesh an old citizen who lives in basavanagudi -:

Ramesh has lived in Basavanagudi his entire life, and his memories of “old Bangalore” are filled with cold mornings, long walks under trees with shade , and evenings that never needed any air conditioning or fan . He spoke warmly about how the city once felt like “a big garden,” where the weather itself was very pleasant enough to support outdoor life and sports . But over time, he has noticed a big shift. In Summers, he says, summers have become extremely hot and the nights no longer are cool the way they used to be . He shared that even in February during his late 20s felt pleasant whereas the heat now in february feels unreal and draining Ramesh also talked about the missing trees iin his neighbourhood and how this contributes to flooding. Areas that never used to flood in the past now bear with very deep water after a heavy shower. One detail he stressed on was the rise in electricity bills because fans and air-coolers are now used almost throughout the year as the weather gets really hot and unpredictable . What touched my heart the most was the weight in his voice when he said, “This is not the Bangalore I grew up in.”

6.2 asha an old citizen who shifted to whitefeild in 2011

Asha moved to whitefeild in 2012 , she describes the area as almost unrecognizable in today's time , she remembers the city with a cool temperature and very few extreme days with light rainfall but since then there has been not a single summer where summers are not as harsh and unbearable especially in high population zone where the area is surrounded by concrete and with very less trees . Asha experienced high risk floods during 2022 and 2024 where roads near her society were filled with water , trees washing away in the water , spending hours waiting due to waterlogging . She also talks about the stress on

people during that event , she also appreciates how the community has adapted and now cleans up the drainage system , checks in on rain and shares rain alerts during extreme weather conditions .

6.3 lived experience vs scientific data

The personal opinions of Ramesh and Asha are close enough to prove the scientific findings in this research. Rising temperatures, warmer nights, increase in floods, and loss of green shade - everything they described matches with IMD records, rainfall trend charts, and urban heat island studies. Their experiences add a huge emotional impact on the data and show how climate change is not just a scientific issue but something which affects people and locals during their day to day life .

7 Impact on local people & urban systems (355 words)

The climate change in Bangalore is not only a scientific problem , it is something which affects residents' feelings every day in their day to day life . One of the most instant effects is on **health**. During summer, cases of heatstrokes , dehydration, and even heat exhaustion have become really common . Locals who once were walking outside in the pleasant weather now avoid going out rather prefer staying in air condition. Older residents and young children are especially at risk. The city's **economy** also gets impacted by climate change, especially during heavy rains and harsh floods. Each flood may damage vehicles, homes, and shops, which forces many families to spend large amounts of money on repairs. Workers who get daily wages , even a single day of flooding can mean a loss in income. Accessing water has become more difficult . Borewells that used to provide reliable groundwater now are mostly dry, especially in areas with rapid urbanisation. Many households are heavily dependent on private water units , which adds up to the financial bills and creates stress during summer months.High temperatures have also left people with no choice but to use air conditioning , leading to higher electricity bills. Families that used fans are now forced to keep cooling units running for long hours just to stay comfortable. This increased demand gives stress and puts pressure on Bangalore's power supply. Transportation suffers as well. Unexpected rainstorms can flood roads and infrastructure , delay public transportation , and trap people for hours. A flood is sometimes enough to bring a logjam.These impacts are partial and not equal . poor neighbourhoods, which often do not have a proper drainage system and green cover, go through worse flooding and heatwaves. Meanwhile, gated societies with generators and better infrastructure can easily sustain small floods . This shows how climate change puts even more stress on the inequalities present in the city .All Together, these effects reveal how climate change puts a great impact on the daily lives of people across Bangalore putting an impact on their health, finances, comfort, and even their safety

8 Adaption strategies for citizens

As Bangalore's climate is continuously changing , many locals are finding ways to adjust their daily lives to stay safe and comfortable . A lot of these tactics originated right from home. Simple steps such as using light curtain colours , adding reflection window frames, or growing indoor plants near windows can help reduce the scorching heat from entering the house .Even switching to LED bulbs and keeping devices unplugged when not in use can reduce house temperatures more than people think .Some families are redoing small aspects of their houses for the temperature to stay down . Cross-ventilation, balconies with shade , and by insulating the ceiling can make rooms noticeably cooler during peak

summer and can save families from scorching heat entering their houses . These small design choices reduce the need to run fans and air-coolers all day.Water conservation is also very important nowadays .. Many households are installing low leveled water taps, fixing leakage quickly, and collecting rainwater in rooftop barrels. In colonies where borewells run dry often, these small fixes help households stay ready during shortages.Communities across Bangalore are also getting together to find fixes . Tree planting workshops . Residents say that even a short row of new trees along a street can make a difference in green shade and comfort . Local groups also are starting to share weather alerts, help clear blocked drainage systems before storms approach , and guide each other during days that are at risk of flooding.Preparing for instant floods has become a part of local life too. Residents keep emergency kits with basic medicines, torches, power banks, water, and a small amount of food. Many have also learned the safest ways to exit their societies when infrastructure gets flooded . These group efforts show that while climate change is bringing new challenges, citizens are also finding ways to stay safe and together.

9 Roadmap for climate resilient urban planning

Bangalore needs a clear path to deal with the climate problems that are going on. This plan should be able to protect people, fix systems, and get the city in place for the hotter , wetter, and more unpredictable weather . Fixing and expanding the rainwater drain network is a top priority . A lot of the drains that are already present are either too narrow or are not functioning , so when it starts to rain heavily, water quickly overflows and starts to get onto the roads. Wider and more well designed drains that are kept with care would greatly reduce flooding.

Another important thing to do is to bring the lake connections back that used to make Bangalore a network of water bodies that helped Bangalore climate stay moderate . Earlier , lakes that were too full would drain into the next one which worked perfectly as a system. This interconnectivity system is no longer there -Reconnecting lakes and restoring natural areas would bring back a system that kept the city running smoothly for hundreds of years.

Buildings and homes also need to be ready for a hotter future. If all roofs needed to be cool, meaning that instead of soaking sunlight they should reflect it , the temperature inside could go down all over the city. For gated communities , adding more parks and protecting the few green covers that are still there would help bring back shade, more types of plants and animals, and better air quality . Another important part is the transportation. Adding more buses and metros , making walking paths less risky and improving public transportation would all contribute to reducing traffic jams.

Water security should also be a top priority. Recycling water which is waste , especially in tech related areas and apartments, can keep borewells from running out of water and make sure there is always enough water, even in hot and dry months. Lastly, putting IoT-based weather sensors in every neighborhood would give people updates on the risks of rain, heat, humidity, and floods. Locals could get ready for a storm with these types of alerts

These steps make up a plan for not only Bangalore that can handle climate change, but also a city that grows in a smart way that protects its people and its nature.

10 Conclusion(271words)

The data from this study makes it very evident that climate change is not very far in Bangalore; it is already impacting people and can be seen . The city's environment has been changing quickly from what older generations used to experience. Temperatures are increasing, rain is harsher, flooding is happening more often, and trees are vanishing . The "Garden City" climate that made Bangalore famous for so long is slowly going away. Summers are getting hotter, nights are getting warmer, and monsoons are becoming more unpredictable .People who live here feel these changes very often and not just on charts. People all over the city are feeling the effects of a changing climate in real and personal ways, from health problems caused by the heat to homes that are flooded, dry borewells, and rise in electricity bills. For a lot of people, the change is also emotional for old residents because they see the city they grew up in not to become something new and harder to get along with .This makes one thing very clear: action must be taken without wasting time . Bangalore can't afford to wait for the situation to get worse . The city needs to grow in such a way that focuses on better water management, better urban planning, protecting trees and cooler building designs with less use of harder materials which absorb heat. The city can still have a future that is climate-resilient, but only if the city decides to go long term over short term . It is now up to every citizen, planner, and government to protect the city's future before the situation goes out of hand.